



Intelligent computing methods in Air Traffic Flow Management

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ARTICLE INFO

Article history:

Received 10 December 2008

Received in revised form 16 June 2009

Accepted 18 June 2009

Keywords:

Air Traffic Flow Management

Meta-Level Control

Multi-agent system

Intelligent computing

Real time traffic synchronization

Reinforcement learning

ABSTRACT

This research presents the application of intelligent computing models in Air Traffic Flow Management (ATFM). Firstly, multi-agent system in grid computing environment is applied to deal with the problem of ATFM synchronization. The developed system consists of software agents, which are implemented in a Computational Grid platform for congestions identification, conflicts resolution and agreements negotiation among the participating airports. A metric criterion, called Agent's Balancing Standard (ABS), is used as a basic index to measure the effectiveness of reducing both the amount of communication among agents and the delay of flights. Secondly, a brief discussion about the Meta-Level Control model is introduced in ATFM issues to improve the efficiency of communication among the agents. The system is developed to analyze the traffic flow information received, identify its importance and process it in the most adequate order.

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1. Introduction

One of the Air Traffic Flow Management (ATFM) issues is a type of Real Time Traffic Synchronization (RTTS) problem (Tanenbaum and Steen, 2002; Tidhar et al., 1992). In most models and systems designed for ATFM, a common critical problem is to achieve simultaneously computing efficiency, safety and synchronization of the decision making process. In a complex environment where the information flow is intensive, to decide the best action for new arriving messages during on-line control is a difficult task, especially in the scenario of message exchange among the agents. This particular problem becomes more important when one considers the development of a distributed computing system (Prevôt, 2002). The tendency is to alter the way of exchanging ATFM operational information (multimedia form) and thus replace verbal language (typical form) with digital message communication.

To manage the air traffic flow efficiently, new methods from Artificial Intelligence such as multi-agent system have been introduced to ATFM successfully. Nguyen-Duc et al. (2003) proposed an approach of cooperation and negotiation among agents using grid computing in a Real Time Traffic Synchronization problem. The objective of the research is to show the applicability of Agent-Based Simulation for Air Traffic Management (ATM) problem. Though the simulation presented was for the EUROCONTROL environment. Collaborative ATFM using multi-agent simulation has also been developed by NASA, where several simple strategies for the airline operations center agents to select routes were evaluated (Wolfe et al., 2007). Tumer and Agogino (2007) also developed a distributed agent-based Air Traffic Flow Management, and their research presented an interesting and updated approach to multi-agent reinforced learning applied to ATM with/without

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human-in-the-loop filtering. Around the same period, a model of a multi-agent system for ATFM in grid computing – ATFMGC was developed by [Dib et al. \(2007\)](#). The ATFMGC consists of software agents, implemented in a Computational Grid platform, for congestions identification, conflicts resolution and negotiation among the Grid participating airports. Agent's Balancing Standard (ABS) was also used as a criterion to balance and measure the amount of communication among agents and to determine the tolerated delay time of the flights.

On the other hand, some problems have emerged due to the intense exchange of messages in ATFMGC when considering more airports. For a fixed ABS, it is impossible to efficiently balance the air traffic flow and the information flow (the message exchange among agents). It is necessary to develop a suitable mechanism to assist the message management and decision making process. Unfortunately, there is no specific analysis for the Information Flow Management (IFM) in the researches by [Nguyen-Duc et al. \(2003\)](#), [Wolfe et al. \(2007\)](#) and [Tumer and Agogino \(2007\)](#). This argument was just mentioned by [Dib et al. \(2007\)](#), but there is no solution for this problem.

The main focuses of this research which may also be considered as the contributions of the paper are: (1) to illustrate the importance of the Information Flow Management in ATFM through the report analyses of the intense message exchange during the experiment from ATFMGC, (2) to present a solution of Information Flow Management in ATFM, as the example of Meta-Level Control to manage the message exchange in cooperation and negotiation in ATFM ([Alves et al., 2008](#); [Crespo et al., 2008](#)).

Meta-Control approach was introduced by [Raja and Lesser \(2003\)](#) in some complicated control problems. It is a process of optimizing an agent's performance by choosing and sequencing domain and control activities. In multi-agent system, Meta-Control approach is useful when the costs or accumulated costs detrimentally affect agent performance ([Raja and Lesser, 2007](#); [Zhang and Makedon, 2005](#)). Also, the Reinforcement Learning ([Sutton and Barto, 1998](#)) is applied in the decision making process in order to acquire experience from the decisions made in a more efficient way.

Using the advantages of the Meta-Level Control approach, a recent research in Brazil analyzed a real situation, the First Integrated Center of Air Defense and Air Traffic Control – CINDACTA I, in Brasilia, which monitors 50% of the air traffic flow in Brazil. In this center, there is no general computing system to manage and synchronize dynamically the traffic and to adequately support the traffic management decision making process. A fusion of Intelligent Computing Methods is being studied to solve this problem though the development of a new Tactical ATFM System ([Alves et al., 2008](#); [Crespo et al., 2008](#)).

The research proposes a system for applying and managing the ATFM restrictive measures adopted by Air Traffic Controller (ATCO – an entity in ATFM process in Brazil) and supervisors working on an Air Traffic Control Center (ACC) – Distributed Decision Making Support System for Air Traffic Flow Management (SISCONFLUX) ([Crespo et al., 2008](#)). A Message Meta Manager (MMM) model, based on the combination of elements from the Meta-Level Control approach and Reinforcement Learning Algorithms (Q-learning and SARSA) ([Russel et al., 1991](#); [Watkins, 1992](#)), is conceived as an important part of the system.

The paper describes two systems: ATFMGC and a part of SISCONFLUX, in order to show the recent researches of intelligent computing methods in ATFM. The following sections are organized in five parts: after this introduction, Section 2 discusses the design of ATFMGC, including the definition of ABS, the design of the agents, the inference process and the grid infrastructure. The ATFM case study using ATFMGC is presented in Section 3. The proposal of SISCONFLUX, which involves Meta-Level Control approach and Reinforcement Learning Algorithms, is illustrated and discussed in Section 4. Section 5 shows the analysis of two models for the ATFM problem. Finally, Section 6 presents the conclusions and the direction of future studies.

2. ATFMGC: a grid-based multi-agent system for ATFM

Artificial intelligence has been successfully employed to solve a variety of ATFM problems. More recently, distributed architectures ([Lhotka, 2006](#)) have also been used as infrastructure for ATFM systems. However, the combination of agents and grid computing for ATFM had not been developed by any known researches. In ATFMGC ([Dib et al., 2007](#)), agents and grid computing were used to reduce conflicts in ATFM. This section presents the development of the grid-based multi-agent system for ATFM.

2.1. Basic considerations

Despite the fact that modern aircraft are equipped with sophisticated electronic equipment, the Air Traffic Flow Management (ATFM), from an operational standpoint, is still heavily relied on human resources rather than automatic systems, considered thus as insipient by the aeronautical community. Airlines, while competing for better schedules and increasing the number of flights, have been provoking congestions in the airports, which is aggravated with the precariousness of the flow management. These congestions, more and more frequent, compel to recurrent practices of ground delay and, more seriously, performing aircraft airborne delay. Having aircrafts waiting for landing authorization, though an usual method, is questionable when analyzed from the perspective of safety, cost and benefit.

The system proposed is aiming to reduce conflicts among flights by analyzing the potential congestion points and solving them before they occur. The scope is, thus, limited to the ATFM pre-tactical and tactical planning. The following constraints in airport terminal area are considered in ATFMGC:

- (a) the arriving flights have absolute priority over departures;
- (b) aircraft are served on a FIFO basis (first in, first out);
- (c) no more than one aircraft is permitted on a particular runway at a time;
- (d) aircraft with special conditions have priority over the others;
- (e) a take-off aircraft may not be released if the subsequent arrival is less than a specified distance from the runway threshold;
- (f) successive departures are spaced at a minimum time separation equal to the departure service time.

Basically, the proposed system analyzes the available capacity of the departure, en-route, and arrival airspace, and takes tactical actions in case the forecast demand exceeds the available capacity. There are four main functions in this system (Weigang et al., 1997):

- (a) to create a schedule for all departing and arriving flights of each airport, trying to maximize the utilization of the runway and the terminal;
- (b) to identify congestion areas with regard to the airspace, aircraft and terminal constraints;
- (c) to negotiate with other agents on the expected traffic flow to and from the airport;
- (d) to re-scheduling the flights, according to the outcome of the negotiation, and communicate the new schedule to all affected agents.

2.2. Definition of ABS

Agent's Balancing Standard – ABS, adapted from Weighted Combined Total Delay (WCTD) (Tidhar et al., 1992), is used to measure the traffic congestion. ABS uses a different delay cost function for the flights landing and take-off. Considering that the objective of ATFMGC is to resolve the Ground Holding Problem, which is to minimize the aircraft in-flight holding. In this research the delay cost function is simplified to multiply every minute of delay by a weight, such as: the ratio of Landing/Take-Off Flights is 70%/30%, the landing weight is chosen as 0.7, and for take-off, the weight is as 0.3.

When an airport is congested, some flights are delayed and the extension of the delay time is proportional to the severity of the congestion. But the flights do not have the same importance and the delay of a flight can have a higher cost than the delay of other flight. Thus, ABS is calculated according to the duration of delay (*minutes_delay*) and its weight (*weight_delay*), as shown in Eq. (1).

$$ABS(t_1, t_2) = \sum_{i=1, n} weight_delay(f_i) \times minutes_delay(f_i) / 10 \quad (1)$$

where $ABS(t_1, t_2)$ is the Agent's Balancing Standard between the instants t_1 and t_2 ; n is the number of flights between the instants t_1 and t_2 ; f_i is the flight in analysis.

In ATFMGC, the system administrator (in real situation, this administrator is the representative of the Air Traffic Flow Management Unit – ATFMU in Brazil) can choose a threshold value for ABS in order to better adapt the system to his/her scenario. A small value for this threshold will typically reduce the average flight delay at the expense of more communication and negotiation among agents. On the other hand, high threshold values will allow a higher average flight delay, incurring in less communication among agents.

The calculation of ABS is carried out in each airport. The flight schedule is arranged for a certain interval of time TS (Time Slice), considering the time of the previously set of departing flights and predicted landing flights. The TS is changeable and can be reduced when the traffic load in the airport is high, or increased when the load is low.

2.3. Negotiation process

Congestion is identified when the total number of flights scheduled for an airport in a given Time Slice (TS) is higher than the airport maximum throughput. When this occurs, flights are retarded or held in airspace. In general, high levels of congestion cause high average of flight delays.

In ATFMGC, congestion is defined in terms of ABS. Whenever the airport's ABS is higher than a given threshold (called acceptable ABS), the airport is said to be congested. When this situation is detected, a negotiation must take place.

Considering that the input flow of an airport is influenced by the output flow of other airports, the agents must interact hours before the congestion actually occurs in order to solve congestions. The agents of each airport negotiate a metric called *average flight delay*, which is derived from ABS. After reaching an agreement, each airport agent must recalculate new ABSs, now considering the flight(s) to be delayed determined by the agreement.

2.4. Design of the agents

In ATFMGC, three types of agents were developed: the *Pre-ATC* agent, the *Tactical Planning (TP)* agent and the *ATC* agent. The first agent is an expert for local air traffic control, which is designed just to simulate the situation for ATFM purposes. The second one is a distributed Knowledge-Based System for Tactical Planning. Its goal is to manage traffic flow over the entire airspace. The *ATC* agent is responsible for managing the traffic flow in and out of the local airport, monitoring any changes and interacting with the external world.

2.4.1. Pre-ATC agent

The *Pre-ATC* agent of ATFMGC is developed according to ATC/ATM rules, which have been defined by the Brazilian Department of Air Space Control – DECEA (Weigang et al., 1997).

This agent has two modules: scheduler and diagnostician. The scheduler is responsible for re-scheduling flights that are in conflict. The diagnostician receives the re-scheduled flight timetable from the scheduler and calculates the ABS. With this ABS, this module contacts the *TP* agent, who will negotiate with agents from other airports.

2.4.2. Tactical Planning agent (TP)

The *Tactical Planning* agent is composed of three modules: *TP* monitor, negotiator and executor. The *TP* monitor is responsible for contacting the scheduler in the *Pre-ATC* agent to perform the re-scheduling of the flight timetable and providing the Time Slice (TS) that should be used in the ABS computation. Also, it is responsible for receiving the re-scheduled flights from the diagnostician and sending them to the negotiator.

The negotiator is the one that actually negotiates the new flight schedules. It receives a re-schedule request and interacts with negotiators from other *TP* agents until an acceptable ABS is reached.

The executor is contacted by the negotiator after an agreement is reached, and then changes the data of the flights that were re-scheduled.

2.4.3. Inference process

The main inference process used in the *Pre-ATC* agent is the same as in (Weigang et al., 1997). The following steps are the inference processes of the *TP* agent:

- (a) The *TP* agent at airport A communicates with the *Pre-ATC* agent at airport B (and any other related airports). The *Pre-ATC* agent at airport B sends the departure flight delay requirements to airport A. These flights depart from A and then fly to B.
- (b) The schedule process of the *TP* agent at airport A re-schedules the departure flights according to the accepted ABS value at A to avoid the airborne delay regarding airport B.
- (c) The diagnosis process checks the new schedule and stores it in the database at airport A if there is no conflict. When any conflict takes place, the Schedule process repeats the work described in (a) and (b) again and again.
- (d) Broadcasts and negotiation generates a message to be sent to the *TP* agents at B and related airports.
- (e) At the same time, ATFMGCs at related airports are also working with the changed schedule database.
- (f) If the system detects conflicts with a related airport, a new schedule is generated at other airports and the information is sent back to airport A. Broadcasts and Negotiation process negotiates with the related airport.
- (g) Evaluation and validation processes verify the actual schedule within a certain time period.
- (h) The *TP* agent at airport A is invoked recursively until there are no more conflicts at airport A and accepted delays in airport B and related airports.

2.5. Grid infrastructure

ATFMGC is proposed to support the sharing and coordinated use of diverse ATFM resources in an air traffic control system. Since the airports are geographically distributed, components of ATFM are operated by distinct controllers, who sometimes operate with differing policies. ATFMGC in every airport is developed as a virtual computing system that is sufficiently integrated to deliver the desired quality of service over all the ATFM.

Using web methods, the controller clients in ATFMGC can perform the computations on their own airport data, and then use the grid computing infrastructure to interact with other controllers. The controller client does not connect directly to the “Final Service” that he/she wants to use, but connects to web service Interface. In this way, we protect the controller clients program from the complexity of grid computing to resolve the ATC/ATM conflicts and insert the flight schedule changes into each airport. As shown in Fig. 1, all the clients communicate through web methods.

In fact, ATFMGC can operate in the airports computing system by communicating with the “Final Services” accordingly. With a simple interface in the client program, ATFMGC allows the users to get computing power in the grid.

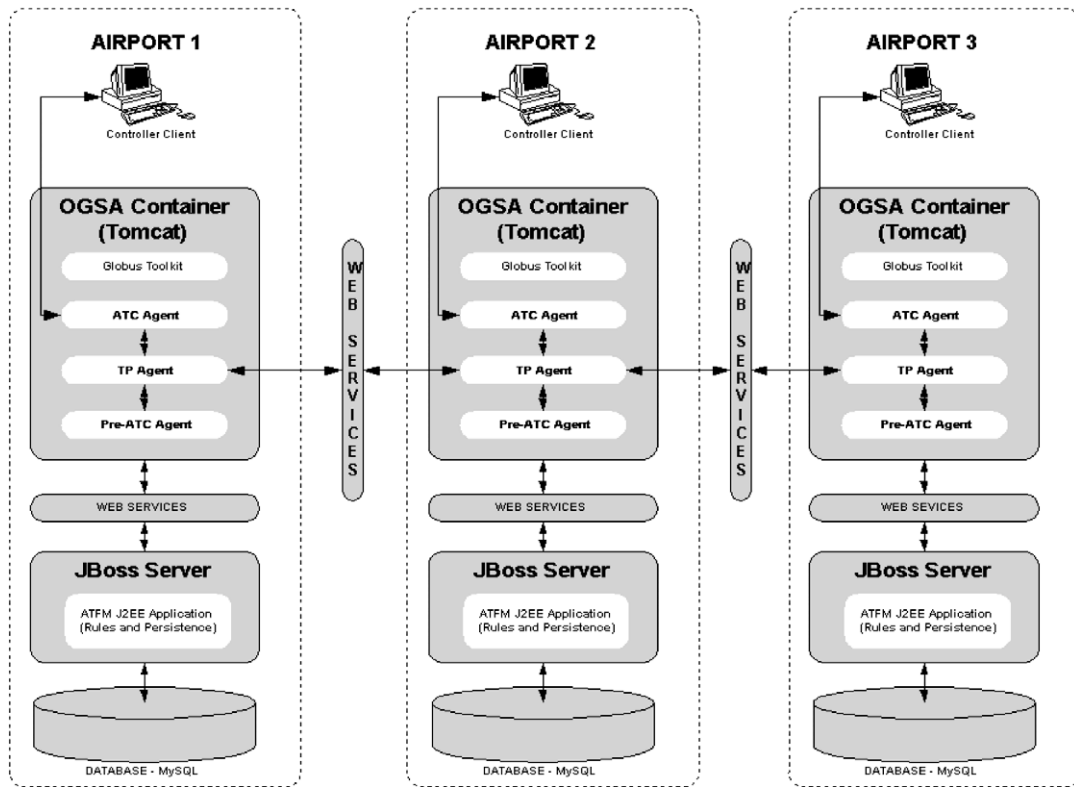


Fig. 1. ATFMGC architecture.

The basic components in the ATFMGC architecture and the relationship among the components are shown in Fig. 1. The proposed ATFMGC architecture consists of five components: Interface, Open Grid Services Architecture (OGSA), Web Services, KBS and Database distributed in a group of airports (Dib et al., 2007).

3. ATFM case study using ATFMGC

Based on the original data of flight schedules on 19, July 2004 (Dib et al., 2007), the airport capacity and the air traffic flow in the four main airports of Brazil (Guarulhos-GRU, Congonhas-CGH, Brasília-BSB, Galeão-GIG) were analyzed.

3.1. Air Traffic Flow in Four Airports

Air traffic flow and airport capacity are different at the four main Brazilian airports. Through this evaluation, we can observe the real traffic situation at GRU airport. We can also demonstrate the capacity of the ATFMGC to rationalize and create flight schedules, considering the minimum time necessary between consecutive flights in landing and in take-off operating procedures. Table 1 shows the legend of some concepts used in this section.

Table 1
Legend of log fields.

Action	Meaning
RTA	Request take-off alteration – message sent by the airport responsible for the take-off, asking the destination airport for an agreement concerning alteration of the landing schedule of a certain flight
OLA	Order of landing alteration – landing alteration request received by the destination airport
RLA	Request landing alteration – message sent by the airport responsible for the landing, asking the airport of origin for an agreement concerning alteration of the take-off schedule of a certain flight
OTA	Order of take-off alteration – take-off alteration request received by the airport of origin
ROA	Reply to order of alteration – message sent by the airport accepting an OLA or OTA received
HE	Time of the event
RH	Requested time in the modification
ST	Status
A/N	Accepted/not accepted

Schematically the messages generated are:

RTA (Take – off) → OLA (Landing) → ROA (Landing/Take – off)

RLA (Landing) → OTA (Take – off) → ROA (Landing/Take – off)

The GRU airport has a capacity of 54 flights per hour (landing and take-off procedures). Using 5 min as a Time Slice (TS) means that the capacity is of five flights in every TS. The real schedule at this airport is 27 flights per hour, but the flights are not well distributed per TS. At some TSs, more than five flights were scheduled. This is the reason for the delay. Analyzing the flight schedule of GRU airport, we can observe that:

- In the actual schedule, with started at 6 h 00 min and ended at 23 h 50 min, the total accumulated delay time was 6 h, in which 55.9% of them are of take-off procedures, with the average delay of 2.1 min per flight, and 31.52% of them are landing procedures, with delays of 1.5 min per flight.
- There is a problem of actual schedule at 20 h 55 min, with four landing flights, and a 21 h 15 min peak of delay for flight take-off procedures. If any meteorological problem occurs, from 20 h 50 min to 21 h 10 min, heavy congestion may take place.

3.2. Simulation

The first task for every TP agent is to get a solution for its conflicts and pass the problems to other airports within the Grid. The TP agents can communicate with others agents to reduce delay, using the ABS to measure the delay situation in every airport and generate a standard balancing.

3.2.1. Application of the ABS concept to the Traffic Flow Management

ABS is locally calculated for each airport, at each TS, and it will indicate traffic and congestion. With a high ABS value, the congestion situation is more complicated. There is a local acceptable ABS based on the traffic capacity of each airport. If the real ABS is higher than the local acceptable ABS, the system needs to communicate with the other airports to negotiate a suitable solution. It is necessary to establish an acceptable ABS to unify the interpretation of the ATFMGC. Five accepted ABS cases were considered in the simulation: 0, 0.75, 1, 1.25 and 2.5. Table 2 shows the result of the simulation.

3.2.1.1. The Accepted ABS is 0. In this case, the system does not allow any delay (even if it is impossible in a real case). For any delay, the system needs to communicate and negotiate between four airports. In this case, a total of 421 communications took place within the system. Just at CGH, the number of negotiation rounds was 218, and it took 8 min to eliminate any delay at this airport (Table 2). Besides, 133 messages were required to eliminate take-off delays of the flights, and 53 messages for landing procedures. The system processed nine messages asking to re-arrange landing procedures and seven to re-arrange take-off procedures, resulting in a total of 16 messages answering to these requests.

Table 2

Number of messages to each airport with accepted abs and execution times.

ABS	Airport	RTA	RLA	OLA	OTA	ROA	Sub total	Total	Time duration
0	GRU	28	1	17	2	19	67	421	6:53
	CGH	133	53	9	7	16	218		8:07
	BSB	64	23	6	4	10	107		8:24
	GIG	9	2	8	1	9	29		7:27
0.75	GRU	11	1	1	1	2	16	98	0:47
	CGH	21	5	6	1	7	40		1:37
	BSB	18	9	1	1	2	31		0:53
	GIG	7	2	0	1	1	11		0:13
1	GRU	0	1	1	0	1	3	33	0:11
	CGH	0	0	6	1	7	14		1:21
	BSB	5	1	1	1	2	10		0:45
	GIG	3	1	0	1	1	6		0:09
1.25	GRU	0	1	0	0	0	1	21	0
	CGH	0	0	4	1	5	10		0:48
	BSB	3	1	1	0	1	6		0:27
	GIG	2	0	0	1	1	4		0:06
2.5	GRU	0	0	0	0	0	0	17	0
	CGH	4	1	2	0	2	9		0:19
	BSB	3	2	0	0	0	5		0:04
	GIG	2	1	0	0	0	3		0

3.2.1.2. The accepted ABS is 0.75. The second execution, which was made with an acceptable ABS equal to 0.75, shows that the agents will accept some conflict in the flights, without contacting the aerodromes of origin (if it is a landing) or the destination aerodrome (if it is a departure). The amount of messages in the Grid for the same period of time of the previous execution dropped from 421 to 98, a reduction of 76.7%, in this case, two messages re-sent to Guarulhos, six to Congonhas, five to Brasilia and none to Galeão.

3.2.1.3. The accepted ABS is 1. In this case, the system allowed a reasonable delay. For any delay, the system needed to communicate and negotiate between four airports. In this case, 33 communications took place within the system. At CGH, the number of negotiation rounds was 14 and it took one and half minutes to eliminate any delay in this airport (Table 2).

3.2.1.4. The accepted ABS is 2.5. This means that the system allowed a considerable delay. In this case, just 17 communications took place. At CGH, there were nine negotiation rounds, and it took 20 s to eliminate any delay in this airport (see Table 2).

The execution times for the complete conflict resolution in the Grid for each airport and acceptable ABS value are shown in Table 3.

3.2.2. Proposal for the management of close source traffic

The proposal for the management of the flow of close source traffic is based on two points. Since a conflict involving two landings is analyzed and resolved 2 h in advance, the system will postpone one of the take-offs whenever possible, that is, when one or both of the flights have not taken off yet. The amount of time postponed will be equal to the time that would be necessary to take the aircraft to its landing slot. It is also assumed that the flight duration is always constant, that is, when a take-off is postponed by “x” minutes, the corresponding landing is also postponed by the same amount of time.

Because the landings have absolute priority over the take-offs, the latter are more subject to delays. Since the take-off is the start of a flight with destination to a certain airport, the ATFMGC, negotiating a new landing schedule with the destination airport, will end up altering the take-off schedule of the departure airport.

Based on the points mentioned, the ATFMGC agents will synchronize all the nodes of the Grid (flights of all the airports) utilizing the Grid services. The method proposed herein is different from the one currently employed, since nowadays the conflict condition is not verified in advance, and the only possible measure to solve it is by holding aircraft in airborne. The expected gain with the synchronization of the flights is the reduction of the holding time in airborne and, consequently, the reduction of the congestion possibility.

In this case study, the method proposed will be used for the synchronization of the flights, as well as the analysis of the behavior of Tactical Planning and Pre-ATC agents. The results of the two methods are compared, and we will see that the method proposed here requires an air holding time that is shorter than the one of the method currently used.

For the execution of this case study, the real time clock of the ATFMGC was started and the agents were called on at 7 h 30 min, for verification of the schedule for the period of 9 h 30 min to 10 h 30 min. The objective was not to show the math-

Table 3

Execution time (in mm:ss.sss).

Acceptable ABS	GRU	CGH	BSB	GIG
0	6:53.408	8:07.354	8:24.220	7:27.628
0.75	0:47.715	1:37.587	0:53.843	0:13.329
1	0:11.896	1:21.377	0:45.431	0:09.794
1.25	0	0:48.478	0:27.106	0:06.800
2.5	0	0:19.453	0:04.366	0:00.360

Table 4

Comparison of airborne delay.

Flight	Airport of origin	Destination airport	Airborne delay (min)	
			Current method	Proposed method
JJ3540	CGH	BSB	1	0
JJ3716	CGH	BSB	0	0
G31834	GIG	BSB	0	0
JJ3704	CGH	BSB	0	0
RG8920	GIG	GRU	2	0
VP4261	GIG	GRU	2	0
RG2329	BSB	GRU	1	0
G31712	GIG	CGH	5	0
RG2635	BSB	CGH	7	0
G31815	BSB	CGH	0	0
JJ3703	BSB	CGH	0	0
VP4281	BSB	GIG	2	0
Total			20	0

ematical precision, but the behavior of the agents. That is why the minimum times of separation were modified, to facilitate the analysis and highlight the conflicts. All the landings at the airports involved were adjusted to have a duration of 2 min, and the departures, 1 min. Table 4 was generated from the comparison of the holding time in airborne of the aircraft involved in this case study, by means of both the current and the proposed method.

The current method for solving the flight conflicts listed in Table 4 requires holding the aircraft in airborne for a total of 20 min, which means an average holding of 1.7 min per flight, verified in the one hour period. The method proposed does not require holding the aircraft in airborne, due to the synchronization of both landing and departure times. The economy of 20 min of flight time, in one hour of flight analysis, is a strong indicator of the benefits of this method's in-depth study.

4. SISCONFLUX

The Distributed Decision Making Support System for Air Traffic Flow Management (SISCONFLUX) is under development by a cooperative program consisted of CINDACTA I and the University of Brasilia (UnB) (Alves et al., 2008; Crespo et al., 2008; Souza et al., 2008). The system proposed aims to list suitable decisions for the users, ATCO, working at an ACC or at an ATF-MU. The decisions will be restrictive measures to be adopted in order to prevent air traffic control sectors saturation. Fig. 2 shows the architecture of SISCONFLUX. Usually, some of the restrictive measures in the Air Traffic Flow Management are: airspeed reduction of flights in route, ground holding, en-route holding and rerouting. In this research, the ATFM restrictive measure considered is ground holding.

Such decisions represent the most adequate solutions for the maintenance of the best possible air traffic flow condition in the Flight Information Region of Brasilia – FIR-BS. SISCONFLUX is also proposed to support the decision making conditioned to some constraints, as the example of prioritized flights, routes or airports. Such prioritized actions may be established in agreement with orientation issued by the national agency for the Air Traffic Flow Management in Brazilian airspace. Among the modules of SISCONFLUX, the MBF and the MAPC were discussed in more detail at Souza et al. (2008) and Ribeiro et al. (2008). The Fig. 2 shows the basic architecture of the system. In this section, MAAD which still is in the developing stage will be further discussed.

4.1. Decision Making Support and Evaluation Module (MAAD)

Basically, the MAAD has to be capable of: (a) working as an interface for the SISCONFLUX and the user; (b) evaluating the air scenario, considering the (foreseen) impact caused by the flow control measures suggested by the MBF, using the criteria of aircraft quantity in the control sectors, amount of generated and possible intrinsic conditions in flow control (prioritization of certain airports); and (c) making the system privilege control measures that project the best scenario, taking into account the evaluation function implemented and the acceptance (or not) of the suggested measures by the user (such as ATCO).

More specifically, the MAAD has the purpose of evaluating a given set of traffic flow control measures, as far as to their adequacy to an existing scenario, using as model for comparison a set of *state-action* type (scenario/control measure taken) and a factor conditioned by the user acceptance level of the new scenario produced. The suggested measure, therefore, will be determined from the analysis of the following three factors: balancing policy suggested by the MBF, set of scenarios produced by the MAPC and experience gained from the options made by the user.

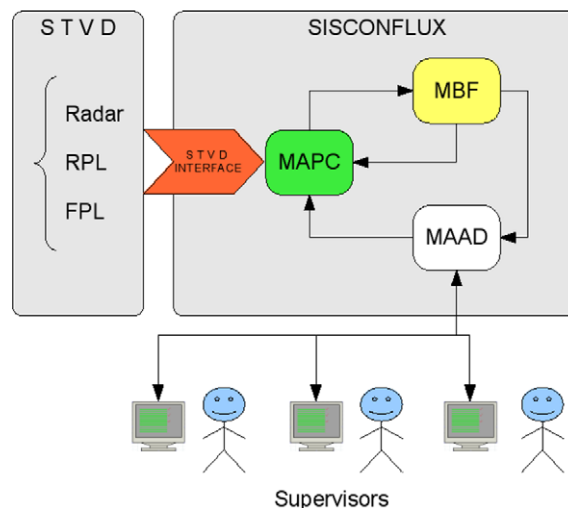


Fig. 2. Architecture of SISCONFLUX, where: SISCONFLUX, The Distributed Decision Making Support System for Air Traffic Flow Management; MAPC, Scenario Forecast Module; MBF, Flow Balancing Module; MAAD, Decision Making Support and Evaluation Module; STVD, System of Treatment and Visualization of Data; RPL, Repetitive Flight Plans; FPL, Eventual Flight Plans.

Within this context, it is inferred that the MAAD has to perform two types of evaluation: the objective evaluation, through the use of quantitative criteria defined by the user; and the subjective evaluation, conditioned by the level of acceptance of the new scenarios created. However, in order for the system to reach a satisfactory convergence level, considering the response time (time complexity) and the users' expectancies (adequate flow adjustment policies), two conditions have to exist, namely: (a) the system has to be capable of learning from the choices made by the users (acceptance/denial of the suggested flow control measures) and (b) the system has to be capable of making a large amount (number to be dimensioned experimentally) of interactions between the modules, and those interactions have the purpose of producing and evaluating, in a simulated way, a variety of flow adjustment policies along with their respective impacts on the scenario.

In order to produce adequate responses to the demand listed above, the MAAD is designed to incorporate intelligent computing methods. In fact, MBF is the primary module responsible for balancing the flow and providing a formal decision to the users. The MBF suggestion consists of ground holding procedures (holding time in minutes) which may (or may not) be adopted by the users and applied to aircraft in elected airports. Based on the users choices, MAAD learns and suggests possible alternatives to adjust traffic flow in the sectors of air traffic control in order to avoid congestion or saturation conditions.

The ground holding procedures are performed taking in consideration some specific conditions and parameters (such as ATC variable sector capacity, airports prioritization, routes prioritization, etc.) which are specified by ATFM authorities and established in the system. In such a way, the MAAD gets the input condition/parameter and adjusts the other modules of the system in order to project a new scenario and restart the operation cycle. MAAD is also conceived to be the human agents interface.

However, the amount of operation cycles necessary to make the system converge in the way previously discussed tends to force the system to exchange a huge number of messages, as verified at ATFMGC. At this point, a discussion about the solution proposed for the message exchange problem, which was pointed out, will take place.

To support the decision making, the technique of Meta-Level Control is modified and applied to the system in order to improve the performance of the communication between the modules. During the control process, the decision is taken based on the experience and the knowledge acquired from the agents with reinforcement learning.

MAAD, as a learning module inserted in the architecture of the SISCONFLUX (Fig. 3) provides adaptation capacity in situations related to the decisions of altering the environment. And the use of reinforcement learning heuristically speeds up the decision making process, where Q-learning and SARSA algorithms are introduced.

4.2. Architecture of the Decision Making Support and Evaluation Module (MAAD)

The model of MAAD is designed as a Meta-Level Control layer that receives the messages, analyzes them in a more appropriate sequence, schedules and prioritizes determined messages. MAAD consists of two main modules: Module of Decision and Control – MDC and Module of Reinforcement Learning – MRL. Meta-Level Control uses a set of parameters in such a way to schedule each message (high utility of the message and maximum stated execution period) and generate a series of other parameters such as: probability of a message to arrive in the entrance list with high utility, the utility of the messages (used to set appointments in the agenda), the execution period for the message that is in the beginning of the agenda and the flow ratio that measures the proportion of messages entering to and leaving from MDC (see Fig. 3).

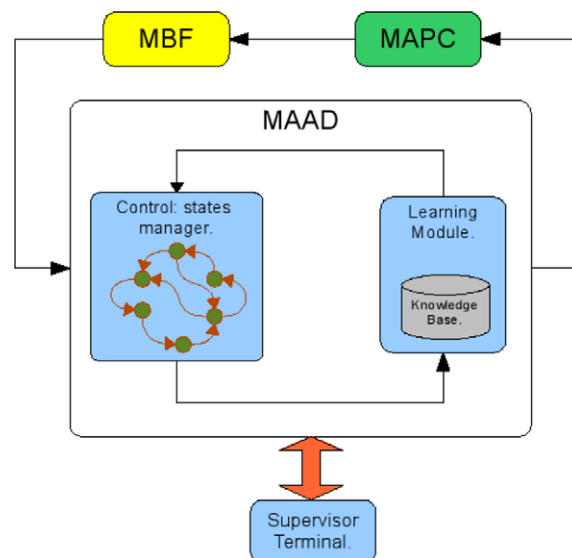


Fig. 3. Information management process within MAAD.

To exchange messages in a distributed system served by a bounded communication channel it is necessary to establish a hierarchy according to system specifications. The attributes defined by the system are attached to the message and the Meta-Level Controller encapsulates and sends it. The destination of the message is also processed by a Meta-Controller within the multi-agent system, which receives the message and analyzes the enclosed attributes to then make the decision in the most appropriate way. The manager in Meta-Level Control can decide among three actions: to set appointments for the message for posterior act of receiving; to transfer the message in the system or to discard it.

The approach of intelligent agent makes use of some aspects of Meta-Level Control such as the use of the Meta-Level parameters attached to the messages in order to make decisions more efficiently without overhead processing and consequent overall system performance constraints.

4.3. Decision and control approach

The decision process is carried out by MDC, which is supported by a Knowledge Base. This base is defined as a part of MRL. The heuristic is used initially in ad hoc form, where for each state of the environment; it suggests an action in accordance with the previous analysis. However, while dealing with a random environment, it does not always suggest the best action first.

In fact a defined heuristic is necessary for an initial situation where the system knows very little about the environment so the agent does not have enough parameters (beginning state). Thus the suggestions change as the agent gets more information about the environment.

The state changes are shared by both modules. MDC is designed for controlling the state changes, managing the entrance and the exit of messages and carrying out the communication among the agents. And MRL incorporates reinforcement learning for better decision making.

Six parameters are used to define a state: p_1, p_2, p_3, p_4, p_5 and p_6 . They are standardized as follows:

- p_1 represents the priority (utility degree) of the arriving messages;
- p_2 represents the arriving messages stated execution period (in entrance list);
- p_3 represents the priority (utility degree) of the appointed set (of messages; in agenda);
- p_4 represents the appointed set stated execution period (in agenda);
- p_5 represents the high utility messages arriving probability;
- p_6 represents the flow ratio of the messages in MAAD.

Each parameter will receive a value: high, average or low (see Fig. 4). When a state has the instance, for example, $(111101011011)_2$, in binary, the values for each parameter are as follows: $p_1 = 11, p_2 = 11, p_3 = 01, p_4 = 01, p_5 = 10, p_6 = 11$, that corresponds the state 3931 in decimal form.

The general process would be very slow due to an intensive use of Reinforcement Learning Algorithms. Thus an alternative approach to speed up the decision making process is necessary. In this sense, the selection between exploitation (of known information) and exploration (of new possible actions) may be influenced by the performance of the agent in the environment and the external factors to the environment.

To evaluate the performance of the current state of the system – MAAD, five parameters are considered to obtain the final value of AvD:

$$AvD = \frac{(3 \times SDS + 2 \times RF + SEL + SAL + SPL)}{8} \quad (2)$$

where

- SDS (Situation of the Host System): this parameter presents situation of the system served by MAAD – it is assigned with weight three in the general evaluation due to the necessity to increase the influence of the host system on the performance evaluation of the MAAD.
- RF (Ratio of the Flow): this parameter is used to measure the flow of the messages in MAAD – it is assigned with weight two due to the necessity to get efficiency on messages exchange in MAAD.
- SEL (Situation of the Entrance Lists): this parameter is used to measure the queue of entering messages and can be reduced or increased when the system is more congested or less congested, respectively.
- SAL (Situation of the Agenda): this parameter is used to measure the Agenda of messages and is reduced or increased when the Agenda in the system is more congested or less congested, respectively.

$$\begin{array}{cccccc} \underbrace{11}_{P_1} & \underbrace{11}_{P_2} & \underbrace{01}_{P_3} & \underbrace{01}_{P_4} & \underbrace{10}_{P_5} & \underbrace{11}_{P_6} \end{array}$$

where: $(11)_2$ = High, $(10)_2$ = Medium, $(01)_2$ = Low and $(00)_2$ = Undefined

Fig. 4. The parameters for state presentation.

- SPL (Situation of the Ready List): this parameter is used to measure the processed messages in the queue and can be reduced or increased when the queue is more congested or less congested, respectively.

4.4. Reinforcement Learning Module – RLM

The agent of reinforcement learning uses two algorithms that are well defined in the literature of reinforcement learning: Q-learning and SARSA (Sutton and Barto, 1998). Four parameters modification strategies in the Q-learning and SARSA algorithms were tested during reinforcement learning simulation: Initial Heuristic (HI), Adaptive Epsilon (EA), Performance based Heuristic (HP) and the combination of them (HI + EA + HP). Eight modifications have been implemented: four modifications for Q-learning and four modifications for the SARSA (Alves et al., 2008).

One of the advantages of the reinforcement learning application is that the agents acquire experience during the interaction process with the environment. In addition it utilizes overall system performance as criteria to verify the performance of the agents in the environment. After analyzing the activities of controllers and supervisors, the order of importance of these activities was established in the computing module, allowing faster and more efficient treatments of more critical situations. At the same time, the controllers and supervisors can get the quantitative impact from their decisions.

The simulation results of the four cases applying two Reinforcement Learning Algorithms (Q-learning and SARSA) had shown the correctness and efficiency of the combination of Meta-Level Control and reinforcement learning in a special problem of ATFM (Alves et al., 2008; Souza et al., 2008). At the stage of research report, the simulation is just a part of the internal message process. Other interesting aspect is the evaluation of changes based on the value of alpha parameter. The learning quality is influenced by the suitable choice of alpha value.

5. Analyses of two systems

5.1. Discussion about the ATFMGC

ATFMGC is a multi-agent system using grid computing to deal with the real situation air traffic flow synchronization problem. In the flight schedules negotiation and synchronization among the airports within the Grid, the schedules converge for a solution without conflicts. In the phase of strategic planning, the ATFMGC can assist the ATFM authorities in the flights planning and scheduling, enabling the identification of conflicts. Its conception allows the verification and real time synchronization of take-off and landings schedules, in all airports related to the flight routes, observing their respective capacities. Agent's Balancing Standard (ABS) is defined to balance the communication among the airport, but it may be used for same kinds of multi-agent system problems as well.

The current concepts of Air Traffic Flow Management encourage the development of ideas that increase the management capacity of the ATFM system through the use of a set of new features that better support the operational dynamics and the current volume of air traffic flow managed by the airspace control centers. The ATFMGC meets those needs, by providing a process of real time synchronization. The combined use of the Multi-Agent and Computational Grid Technologies represents a significant advance in the process of close source ATFM.

In the phase of strategic planning, the ATFMGC is able to help the air traffic flow managers in the planning and elaboration of the flight scheduling, enabling the identification and reduction of the overload, as well as the detection and resolution of conflicts. Its structural components are capable of performing the verification and synchronization in real time to the take-off and landing schedules of all the airports along the routes of the flights, considering the use of each airport's capacity and the air traffic flow.

This tool also allows the customization of the aeronautical infrastructure usage and values for minimum separation times specific for each airport, thus generating a flight Schedule free of conflicts and congestions, the ultimate goal of all the air traffic controllers and air traffic flow managers, in order to provide a better service to the users of the air transport, as well as increasing the safety of the flights and reducing the operational costs of the airline companies.

In the phase of tactical planning, the ATFMGC implements an approach for the Air Traffic Flow Management with applicability in the solving of close source flights conflicts, enabling the re-planning, in real time, of the flight schedules of the airports participating in the Grid. It is a set of coordinate actions performed by a triad of software agents, acting in real time to solve schedule conflicts between the flights, resulting in a significant gain in terms of safety and savings when compared to the current method.

The ATFMGC system has been implemented in a Computational Grid environment, with the use of four computers, representing the four busiest Brazilian airports in terms of passenger number: Guarulhos, Congonhas, Brasília and Galeão. Also, a data base, which was being fed with real information of the flights planning, was generated to support the experiments being conducted. The four "airports" were configured for the execution of Air Traffic Flow Management operations simulations, under various loads of messages, which made it possible to demonstrate the Grid performance.

Furthermore, the ATFMGC system and its Grid platform were built with free software, meaning that it has low costs of development and can be easily implemented in the real world. It was demonstrated that the air traffic flow regulation through the use of the ATFMGC significantly reduces 20 min of the holding time in one hour of flight analysis, by comparing the values obtained in the study cases to the ones generated by the current procedures utilized by the air traffic controllers.

Although the ATFMGC has been designed to support the tactical planning, it has potential to perform with safety and efficiency in the phases of monitoring and control directed to the synchronization in real time, on account of its conception, capacity and implementation platform.

The utilization of coordinate multi-agents combined with the computational Grid technology for the Air Traffic Flow Management and synchronization in real time has proved rather promising. The results obtained met the expectations, showing that it is an interesting topic to be explored in future researches due to the significant gain in terms of air traffic safety and prospective savings for the airline companies and their users.

As mentioned above, ATFMGC was developed to address a fundamental problem: Traffic Flow Management. Using a multi-agent architecture to resolve the Ground Holding Problem – GHP, considering the complex activity of synchronization of take-offs (both airports of origin and destination) and landings (both airports of origin and destination) to manage the air traffic flow.

During the traffic flow management in ATFMGC, Agent's Balancing Standard (ABS) was also used as a criterion to balance and measure both the amount of communication among agents (i.e. airports) and the tolerated delay of the flights. This can be named as Information Flow Management (IFM). Some problems emerged on account of the intense messages exchange when using ATFMGC applied to more than 10 airports. In fact, for a fixed ABS, it is impossible to efficiently balance the communication. The intense message exchange among the agents (airports) was required in order to achieve the adjustment of landing and take-off schedules (synchronization of traffic flow) for no congestion for both air traffic flow and information. There are no previous references to this problem. At this point, it can be verified from Table 3 the importance of Information Flow Management in ATFM, concerning the exchange of messages among agents, seeing that the time of execution of the synchronization is significantly impacted by the number of messages exchanged by agents.

On the other hand, ATFMGC does not provide a solution to the problem presented, namely, intense message exchange, but it alerts the need to develop adequate tools to solve it. This is the main point of developing SISCONFLUX, which uses artificial intelligence to manage (synchronize, prioritize, optimize) the flow of messages between agents, so that the management of traffic flow occurs within an acceptable execution time.

5.2. Discussion about the SISCONFLUX

The number of variables and factors involved in the Air Traffic Flow Management, with special highlight for the indispensable analysis of the impacts resulting from the flow adjustment policies devised, requires the presence of a supporting computational tool in order to have an efficient and reliable decision making process.

Nevertheless, the ATFM process constitutes a domain where the complexity inhibits the adoption of fully automated methods and/or tools, especially when one considers its intrinsic relationship with the air traffic control, characterized as a critical activity performed in real time.

Thus, a viable solution for such a dilemma will necessarily pass through the conception of a system containing at least the following two agents: the non-human agent (computational tool), which suggests actions (flow adjustment or flow control policies) relative to a certain scenario; and the human agent, who has the prerogative of accepting or not accepting such suggestions.

In this context, it is possible to infer that an ATFM computational tool, which may be classified as a decision supporting system, will have its efficiency strongly conditioned by the way the man–system interaction is modeled. More specifically, it can be said that the higher capability of the system to accommodate the expectations of the human agent, the more efficient the system will be, considering criteria established by the human agent and his/her experience, in the case where he/she is the ATFM specialist and user of the system.

However, a crucial problem emerges here, which is finding how to make the system produce adjustment policies that aggregate some level of convergence relative to the human agent's expectations. The Artificial Intelligence has the solution: provide the system with learning ability, acquiring experience based on the set defined by the scenario, by the flow adjustment policies that the system produced, and by the action taken by the human agent (filter).

6. Conclusions and further research

By describing the design of the two systems, ATFMGC and SISCONFLUX, it was shown that two kinds of traffic flows should be considered for an ATFM system: the flight flow itself (Air Traffic Flow Management) and the communication message flow (Information Flow Management). This consideration may be seen as an innovation for modern ATFM in the near future.

The researches of multi-agent systems and Computational Grid application in the Air Traffic Flow Management domain are still in developing stage. The in-depth study of the topics listed below seems promising for future researches.

- (1) In the tests, it was verified that the flow of messages increased considerably at certain times, where an agent made and received various requests simultaneously. In the model developed, this problem was initially tackled with the complete execution of the cycle of conflict analysis, but this procedure generated extra processing that can be potentially prevented. If more airports are incorporated in the Grid, this procedure may prove inadequate. Two aspects, therefore,

should be studied. The first one is related to the prioritization of the requests accommodation, since no distinction was made among the airports. The second one is related to the management queues for accommodation requests, possibly solved by creating a proper protocol for the negotiation between the agents in this environment.

- (2) The Computational Grid solution adopted, based on the Open Grid Service Architecture – OGSA, worked well and had a reasonably good performance in the four airports simulation studied. Although the Grid converged to an overload situation, the convergence time was not satisfactory. The service Grid is based on the Web Services, which work with the XML standard. This type of technology is known to utilize excessive “parser” to treat the xml, something that can affect negatively the system processing. For the utilization of this architecture in a real environment, it would be necessary to research with the use of more computers in the same node of the Grid to evaluate the multi-thread performance, or research another type of Grid platform for the implementation of the multi-agents.
- (3) In the model implemented, the flights were considered as ideal, with constant durations and procedure execution times. Factors, such as winds, turbulence, etc., which influence on the duration of the flights, were disregarded. The consideration of these factors in the forecast of the estimated landing time would represent an interesting progress in the utilization of data in real time. In the near future, the non-constant flight duration should be considered.
- (4) The interconnection of the ATFMGC with the data received from a radar station, together with mathematical models concerning the influence of the winds, would certainly represent the acme in the synchronization of air traffic flow data in real time.

In the development of SISCONFLUX, the integration of decision making supporting procedures among all modules of SISCONFLUX will be pursued in order to more effectively assist the ATCO and supervisors activities. It is intended to consider a heuristic procedure in MAAD to improve the performance, reflecting quantitative criteria of the real life ATCO and supervisors performance in CINDACTA I. It is also interesting to introduce the solution of multi-agent system in SISCONFLUX, or better adapt the Agent's Balancing Standard (ABS).

Acknowledgements

This research is partially supported by Brazilian National Council for Scientific and Technological Development – CNPq (Procs. 306065/2004-5 and 485940/07-8) and ATECH Tecnologias Críticas, Brazil. Special thanks to the reviewers, Dr. Matthew Karlaftis, Luiz Nelson Marcelino Dias and Daniel Lezhi Li for their editing.

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